

Biodegradable polymers for marine antibiofouling: Poly(ϵ -caprolactone)/poly(butylene succinate) blend as controlled release system of organic antifoulant

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ARTICLE INFO

Article history:

Received 6 September 2015

Received in revised form

31 January 2016

Accepted 5 March 2016

Available online 9 March 2016

Keywords:

Poly(ϵ -caprolactone)

Poly(butylene succinate)

Biodegradation

Controlled release

Biofouling

ABSTRACT

Polymer carrier is critical for the controlled release of antifoulant in marine anti-biofouling. We have prepared biodegradable polymer blend consisting of poly(ϵ -caprolactone) (PCL) and poly(butylene succinate) (PBS) and used it as carrier of organic antifoulant (4,5-dichloro-2-octyl-isothiazolone). Polarizing optical microscope (POM) and differential scanning calorimetry (DSC) measurements show that the spherulite size of PCL decreases and its crystallinity slightly varies as PBS fraction increases leading the interfacial regions between PCL and PBS spherulites to increase in the total area. Our studies reveal that such amorphous interfacial regions allow the blend to degrade at a constant rate and the antifoulants to be uniformly dispersed in the blend. The antifoulants are released at a constant rate as the blend degrades in marine environment, which can be modulated by the blend composition. Marine field tests demonstrate that such a system has good marine anti-biofouling performance.

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1. Introduction

Marine biofouling is a widespread problem affecting marine industries [1]. Tributyltin-containing coatings which were effective against marine biofouling were banned due to their harmfulness to marine environments, there is a need for environment-friendly antifouling systems [2]. As alternatives, synthetic and natural organic antifoulants [3,4], polymeric materials such as zwitterionic polymers [5,6], functional polymer brushes [7–9], low surface energy polymers [10,11] and materials with slippery liquid-infused surfaces [12,13] and microstructured surfaces [14,15] have been proposed. An antifouling system typically contains a polymeric matrix carrier and antifouling agent(s), whose controlled release is critical for its performance and service life.

Biodegradable polymers are promising anti-biofouling materials

since their decomposition and erosion by seawater or enzymes can lead to a self-polishing surface [16–19]. On the other hand, they can serve as carrier and release system of antifoulants. Poly(ϵ -caprolactone) (PCL) and its copolymers were used as the carrier of antifoulants. However, they have limited antifouling ability when their degradation is slow [20–22]. We have developed degradable antifouling systems via physical and chemical modification of PCL. With the controlled release of organic antifoulant, the systems exhibit excellent antifouling performance [23–26]. Certainly, the degradation rate of the polymer is critical since it determines not only the erosion rate of the surface but also the release rate of antifoulants.

Poly(butylene succinate) (PBS) is another well known biodegradable polymer, whose monomers are made from renewable resources [27–29]. In comparison with PCL, PBS has higher density of ester groups in the main chain. PCL and PBS homopolymers do not have good performance in anti-biofouling because of their slow degradation due to the high crystallinity and big spherulite [26,30]. However, the crystallinity of PCL could be reduced by blending with PBS in principle, which would make the degradation and the release of antifoulants faster. Actually, the degradation rate can be modulated by the blend composition. In the present study, we have

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prepared marine anti-biofouling material based on PCL/PBS blend which serves as carrier and controlled release system for a relatively environmentally friendly antifoulant (4,5-dichloro-2-octylisothiazolone) which rapidly degrades in natural seawater without bioaccumulate [2,31–33]. We have examined the degradation of the blend, the release of antifoulant and the antifouling properties of the system. Our aim is to correlate the antifouling performance and the composition of biodegradable blend and develop high performance marine antifouling system.

2. Experimental

2.1. Materials

Poly(ϵ -caprolactone) (PCL, $M_w = 3.7 \times 10^4$ g/mol) was from Perstorp. Poly(butylene succinate) (PBS, $M_w = 4.3 \times 10^4$ g/mol) was supplied by Anqing Hexing Chemical Co., Ltd. 4,5-dichloro-2-octylisothiazolone (DCOIT) was presented by Thankful Chemical Co. (Guangzhou). Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2013) [34]. All other reagents were used as received.

2.2. Preparation of PCL/PBS blend

The blend was prepared by solution mixing. Typically, PCL (7.2 g, 72.0 wt%) and PBS (1.8 g, 18.0 wt%) were dissolved in chloroform (CHCl_3) under vigorous stirring at 55 °C. Subsequently, DCOIT (1.0 g, 10.0 wt%) was added into the PCL/PBS solution and continuously stirred for another 2 h. The solution was casted on substrate to form a polymer film. Other blends with different compositions were prepared using the same procedure. The thickness of prepared films is ~ 400 μm . Note that spray coating instead of solution casting is used in industries. The two procedures can lead to different surface topographies. Yet, for biodegradable polymer antifouling coating, the surface topography has slight effect on the antifouling performance [23–26]. This study can provide useful information for industries.

2.3. Determination of crystallinity and crystal structure

Thermal behavior was analyzed on a Mettler-Toledo DSC1 STAR system with a FRS5 sensor under a nitrogen flow of 50 ml/min. After a sample (~ 5 mg) was first melted at 150 °C for 5 min to eliminate thermal history, it was cooled to -90 °C at a rate of 10 °C/min, kept at -90 °C for 5 min, and then reheated to 150 °C at a rate of 20 °C/min. All DSC traces were from the second heating to minimize effect of thermal history. The glass transition temperature (T_g) was taken as the midpoint of the transition. The melting temperature (T_m) and melting enthalpy (ΔH_m) related to crystallization of PCL phases were taken as the position of the peak and the area of melting endotherm in the heating run, respectively. The crystallinity index of PCL (X_c) was calculated from $\Delta H_m/w\Delta H_0$, where w is the weight fraction of PCL in the blend and $\Delta H_0 = 136$ J/g is the melting enthalpy for PCL with 100% crystallinity [35].

Polarized optical microscopic (POM) observation was performed on a ZEISS Axio Scope A1 equipped with a digital camera. The film was prepared by spin-casting on a spin coater (CHEMAT, KW-400) at 4000 rpm in air using CHCl_3 (25 wt%) as solvent. After the film was melted at 140 °C, it was cooled to 90 °C and kept at least 12 h, and then kept at 40 °C for 24 h.

2.4. Hydrolytic degradation test

The hydrolytic degradation test was performed on PCL/PBS blend without antifoulant in ASW at 25 °C. Since the hydrolytic

degradation occurs on the surface, we used the weighing method [23–26]. The blend film was prepared on epoxy resin panel (20×20 mm²) via solution-casting. The weighted sample (W_0) together with its panel was incubated in a tank of ASW that was renewed every two weeks. At a certain period of time, the panel was taken out, rinsed with deionized water three times, and dried in vacuum freeze dryer until constant weight (W_t) was obtained. The mass loss was designated as $(W_t - W_0)/\text{test area}$. For each sample, three coated panels were prepared and measured, and each data point was averaged over three successive and consistent measurements.

2.5. Determination of antifoulant release rate

The distribution of DCOIT on the blend surface or in the cross-section was measured in terms of chlorine element by using a scanning electron microscope (NOVA NANOSEM 430) equipped with an energy dispersive X-ray spectrometer (EDS) in Smart Map mode at 20.0 kV. Each sample was sputter-coated with platinum for enhanced conductivity.

DCOIT concentration was measured to determine its release rate by use of UV–vis spectrophotometer under a static condition [36]. Typically, a blend film was prepared on an epoxy resin panel (40×40 mm²), and then the panel was immersed in ASW. At a certain period of time, the panel was transferred from the holding tank to an individual measuring container. After 24 h immersion, 100 ml of seawater was taken from 250 ml of seawater in the individual measuring container into a 250 ml glass separating funnel, and it was extracted with 10 ml of hexane, and then the concentration of DCOIT was determined quantitatively by an Agilent Cary 60 UV–vis spectrophotometer at wavelength 290 nm. The release rate of DCOIT from the film was calculated from the DCOIT concentration in ASW using the following equation

$$R = (C \times 250 \times D)/A \quad (1)$$

where R is the release rate of DCOIT ($\mu\text{g cm}^{-2} \text{d}^{-1}$), C is the DCOIT concentration in ASW ($\mu\text{g ml}^{-1}$), D is the immersion time in the individual measuring container (one day) and A is the surface area of coated film (16 cm²). The cumulative release of DCOIT was calculated as follows

$$R_{0,x} = \sum \frac{(R_i + R_j)}{2}(j - i) \quad (2)$$

where $R_{0,x}$ is the cumulative release from Day 0 to Day x ($\mu\text{g cm}^{-2}$), i and j are time elapsed (days) since the start of the trial, for example, 0 and 1, 1 and 7, 7 and 21 days, and so forth, respectively. R_i and R_j are the release rates ($\mu\text{g cm}^{-2} \text{d}^{-1}$) of DCOIT from Day 0 to Day x , for example, on Days 0 and 1, Days 1 and 7, Days 7 and 21, and so forth, respectively, where Day 0 is included, the release rate on Day 0 (R_0) is taken as 0 $\mu\text{g cm}^{-2} \text{d}^{-1}$ of DCOIT.

2.6. Marine field tests

The field tests were performed from September 2014 to January 2015 in South China Sea (22°33'N, 114°31'E) in China, where the temperature of surface water ranges from 25 to 28 °C and the salinity is about 35‰. The samples applied onto the glass fiber reinforced epoxy resin panels ($100 \times 100 \times 3$ mm³) were lowered into seawater at a depth of 1 m. After a certain period of time, the panels were taken out of the sea, carefully washed with seawater and photographed, and then placed back in the sea to continue the test. The antifouling activity was evaluated according to ASTM D6990-05 (2011) [37]. The range of the fouling rating (FR) was

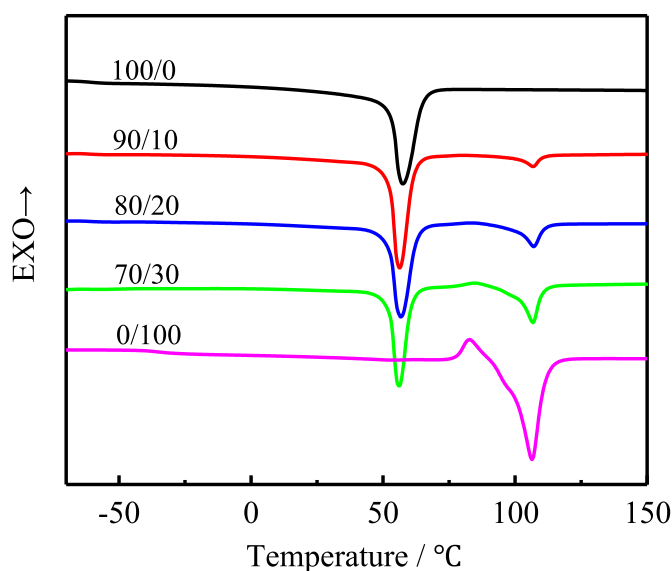


Fig. 1. DSC curves of PCL/PBS blends.

0–100. FR for a coating free from biofouling settlement was recorded as 100. FR of each coating was obtained by deducting from 100, depending on the percentage of area covered by macrofouling, that is, the fouling rating essentially reflects the nonfouled area. There, the biofouling pressure mainly comes from crustacea, anneliada, mollusca, bryozoa and algae [38]. Biofouling attachment within 1 cm from the edge was not counted.

3. Results and discussion

DSC curves of PCL/PBS blends are shown in Fig. 1. All PCL/PBS blends exhibit a T_g (about -60°C) the same as that of PCL homopolymer, whereas T_g corresponding to PBS is not apparent because of its low fraction (Table 1). On the other hand, in comparison with PCL having a single endothermic peak with $T_m = 57.5^\circ\text{C}$ and $\Delta H_m = 78.1\text{ J/g}$, each PCL/PBS blend exhibits two melting peaks corresponding to the melting of PCL and PBS, respectively. Namely, the blending does not change T_m of PCL or PBS. As shown in Table 1, the blending also slightly changes the crystallinity of PCL. All the facts indicate PCL and PBS are immiscible.

The spherulitic morphology of PCL/PBS blend was observed by POM (Fig. 2). The diameter of PCL spherulites is $\sim 100\text{ }\mu\text{m}$ (Fig. 2a). After PBS is introduced, the size of PCL spherulite decreases (Fig. 2b–d). This is because the surrounding confinement of PBS crystals inhibits the growth of PCL crystal [39]. Since PCL and PBS do not cocrystallize, the interfacial area is amorphous. As PBS content increases, the total interfacial area increases. Note that we did not observe macroscopic phase separation by POM. The interfacial regions are probably uniformly distributed in the blend at least at a

micrometer level.

We examined the degradation of PCL/PBS blend. The time dependence of the mass loss of PCL/PBS blends with different compositions in ASW at 25°C is included in Fig. 3. Each film exhibits a relatively rapid mass loss at the initial stage. This is probably caused by a quick release of some low molar mass components [40,41]. After 7 days, the mass loss is almost linear with time, indicating a degradation with a constant rate. As PBS content increase, the degradation rate increase from 1.9 to $3.3\text{ }\mu\text{g cm}^{-2}\text{ d}^{-1}$. This is because the blending leads the crystallinity and the spherulite size to decrease. As we know, the crystalline regions are denser than the amorphous regions, it is much more difficult for water, DCOIT or other small molecules to penetrate into or out of the crystalline domains. The introduction of PBS reduces the PCL spherulite size and increases the total area of the interfacial regions between PCL and PBS spherulites. The amorphous interfacial region is easier for small molecules to penetrate. As a result, the degradation rate increases as the spherulite size decreases. Similar phenomenon was also observed before [26,42,43]. On the other hand, the degradation rate is also affected by the density of ester groups which are the potentially degradable sites [44,45]. The blends with higher PBS content has a higher ester group density and hence a higher degradation rate. Note that the blends with PBS content higher than 30 wt% were not examined because of their poor adhesion to the substrate. All the blends with PBS content lower than 30 wt% exhibit adhesion strength above 1.0 MPa, sufficient for binding of coating to the substrate in marine environment [46].

The cumulative release of DCOIT from PCL/PBS blend in a certain period of time is included in Fig. 4. For each film, the amount of released DCOIT is linearly related to time with a correlation coefficient >0.99 , indicating the release has a constant rate once the blend composition is certain. Moreover, the slope increases with PBS fraction. This is because the degradation rate increases and the DCOIT release rate increases as PBS fraction increases.

The controlled release of antifoulant determines the duration and antifouling performance of an anti-biofouling system. For a system consisting of insoluble resin or soluble matrix (e.g., rosin based coatings), the antifoulant usually has a quick release at the initial stage and then the rate exponentially decreases with time, which is not favorable to the antifouling and duration [47]. To understand why PCL/PBS blend has a constant release rate of DCOIT, we studied the DCOIT distribution in the blend. DCOIT distributions on the blend surface and in the cross-section before and after immersion in ASW are included in Fig. 5. Clearly, DCOIT molecules are uniformly dispersed in the blends. After immersion for about four months, DCOIT molecules still have a uniform distribution. This is because DCOIT is hydrophobic with a water solubility of $5\text{ }\mu\text{g ml}^{-1}$ [32] and it does not readily diffuse into seawater. As the polymer blend degrades with a constant rate, DCOIT are gradually released. That is why the release has a constant rate. Namely, the release of DCOIT from PCL/PBS blend is controlled.

Since the marine environment is complicated where marine

Table 1
Characterization data of PCL/PBS blends.

PCL/PBS	T_g (PCL) ^a	T_m (PCL) ^a ($^\circ\text{C}$)	T_m (PBS) ^a ($^\circ\text{C}$)	ΔH_m (PCL) ^a (J/g)	X_c (PCL) ^b (%)
100/0	−60.3	57.5	—	78.1	57.5
90/10	−60.4	56.4	106.9	70.0	57.2
80/20	−60.2	56.9	107.1	59.9	55.0
70/30	−59.7	56.2	106.8	51.9	54.6
0/100	—	—	106.4	—	—

^a Determined by DSC.

^b Calculated from $X_c(\%) = \Delta H_m / w \Delta H_0 \times 100$, where w is the weight fraction of PCL in the blend and ΔH_0 for 100% crystalline PCL = 136 J/g [35].

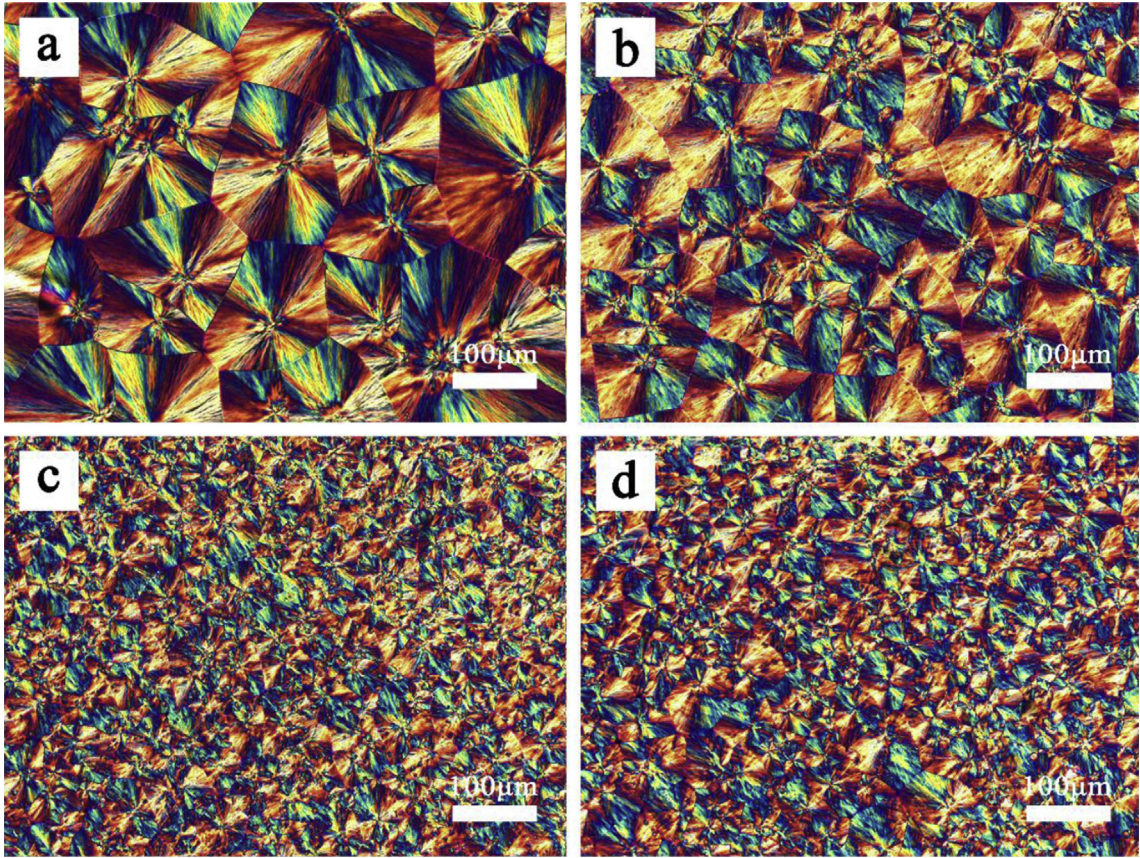


Fig. 2. POM images of PCL/PBS blends. (a) 100/0, (b) 90/10, (c) 80/20, (d) 70/30. The scale bar is 100 μm.

fouling varies with the seas and the seasons, we used marine field test to evaluate the antifouling performance of the polymers. The typical images of tested panels after immersing in the sea as a function of time are included in Fig. 6. The corresponding fouling rating for four months is shown in Fig. 7. For a control panel with epoxy surface, algae, bryozoans and calcarina already grow on the surface in one month, and the surface is completely covered by the

biofouling in four months, indicating a heavy fouling pressure. For panels coated with PCL without DCOIT, it is not surprised that some marine organisms are located on the surface in one month, and the surface is heavily fouled in four months because PCL with a low degradation rate has poor self-polishing ability. However, much less fouling is observed on the surface of the panel coated with PCL/PBS blend without DCOIT in one month. This is because the

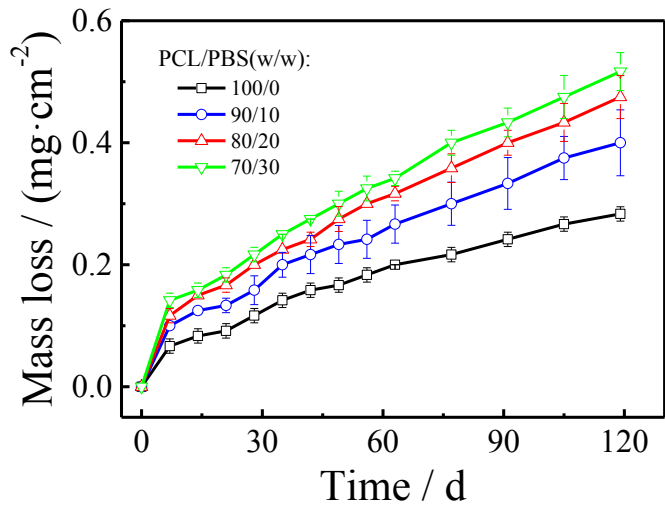


Fig. 3. Time dependence of mass loss of PCL/PBS blend in ASW at 25 °C.

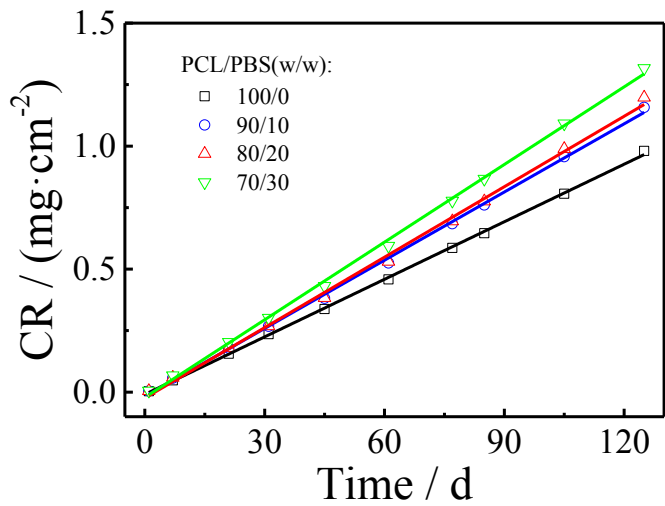


Fig. 4. Time dependence of the cumulative release (CR) of DCOIT from PCL/PBS blend.

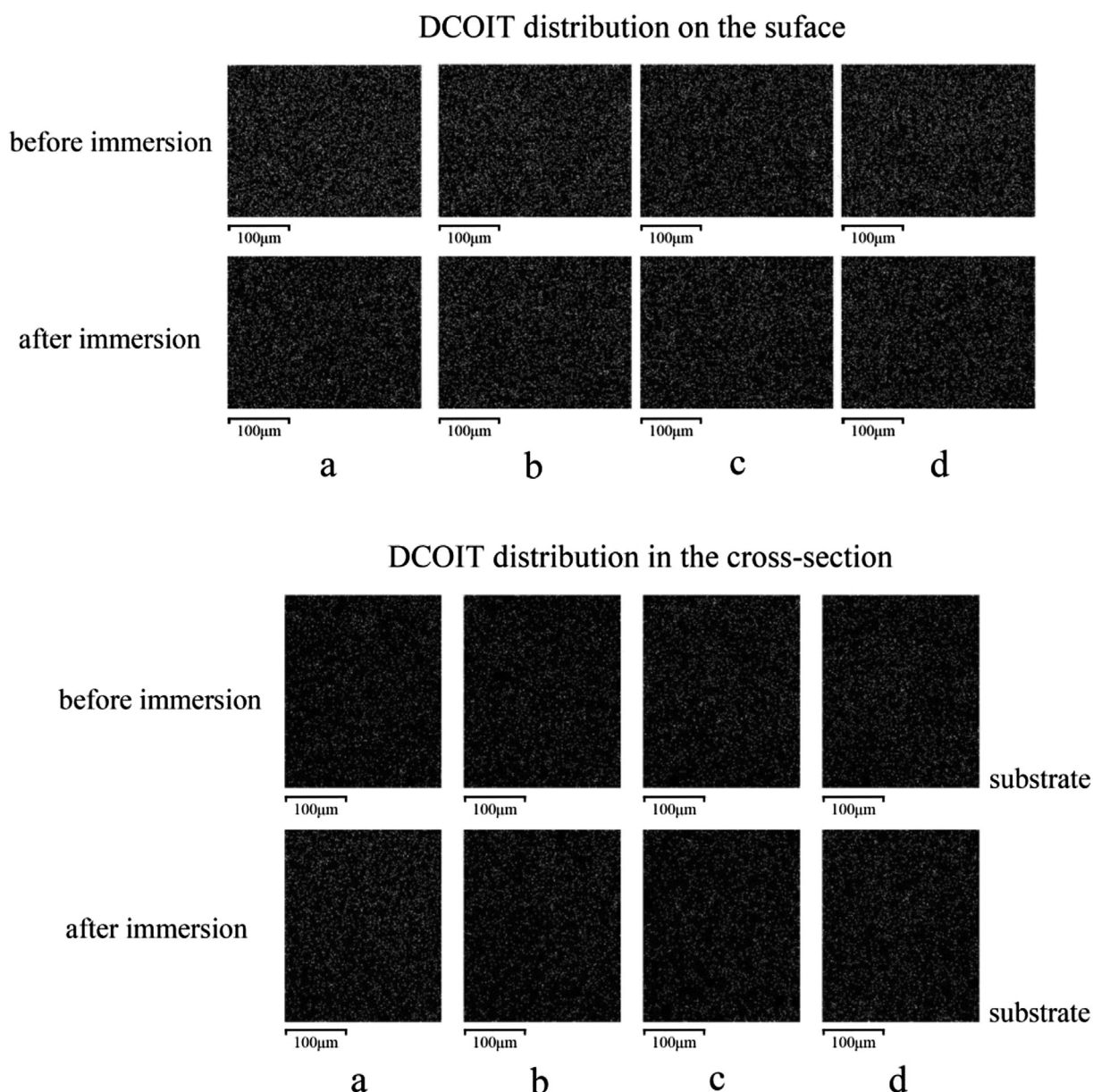


Fig. 5. DCOIT distribution on the surface or in the cross-section of PCL/PBS blend before and after immersion in ASW for four months. (a) PCL/DCOIT = 90/10, (b) PCL/PBS/DCOIT = 81/9/10, (c) PCL/PBS/DCOIT = 72/18/10, (d) PCL/PBS/DCOIT = 63/27/10.

blend degrades faster leading to a more quickly changeable surface that can prevent biofouling from adhesion to some degree. Accordingly, the polymer degradation is constructive to improve the anti-biofouling ability of the system. For panel coated with PCL/DCOIT, the surface is not fouled in one month but has some fouling on the surface in four months. This is understandable because PCL/DCOIT system with slow degradation rate and slow DCOIT release rate cannot inhibit the heavy biofouling. For panel coated with PCL/PBS/DCOIT, the surface is clean in four months. The blending of PBS with PCL modulates the degradation rate to a proper level so that the antifoulants are released at a reasonable rate. As a result, the system exhibits a good anti-biofouling performance and longer duration. Note that the degradation rate and the antifoulant release rate increase upon water flow. Namely, the system is expected to exhibit better antifouling performance on dynamic conditions. In short, a system with reasonable

degradation rate and controlled release of antifoulants would have good anti-biofouling ability.

4. Conclusion

We have explored the possibility to use biodegradable poly(ϵ -caprolactone) (PCL)/poly(butylene succinate) (PBS) blend in marine anti-biofouling. The blending of PBS with PCL reduces the spherulite size of PCL and increases the total area of the amorphous interfacial regions between PCL and PBS spherulites. The degradation rate increases with PBS fraction. The antifoulants are released at a constant rate once PBS fraction is fixed and it can be controlled by changing the blend composition. PCL/PBS/antifoulant system shows good antifouling ability and long duration. The system is expected to be applied in marine antifouling.

Time / month

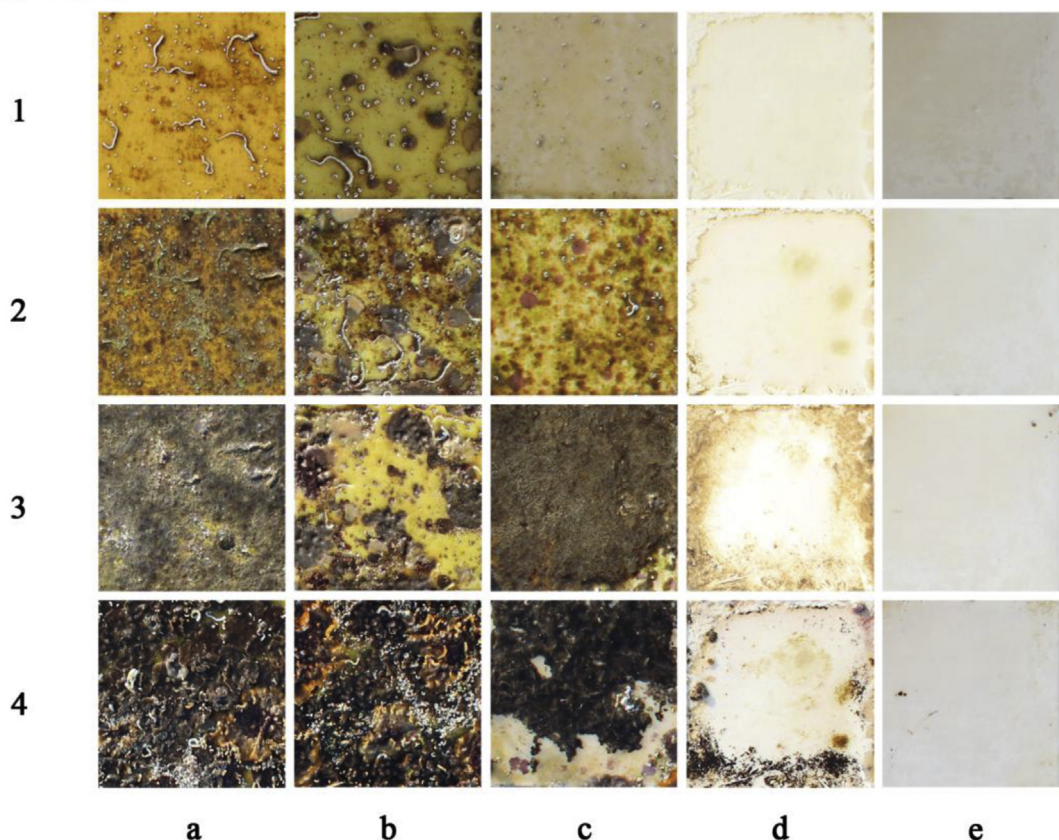


Fig. 6. Images of panels after immersion in the sea as a function of time. (a) control panel without coating, (b) panel coated with PCL, (c) panel coated with PCL/PBS = 80/20, (d) panel coated with PCL/DCOIT = 90/10, (e) panel coated with PCL/PBS/DCOIT = 72/18/10.

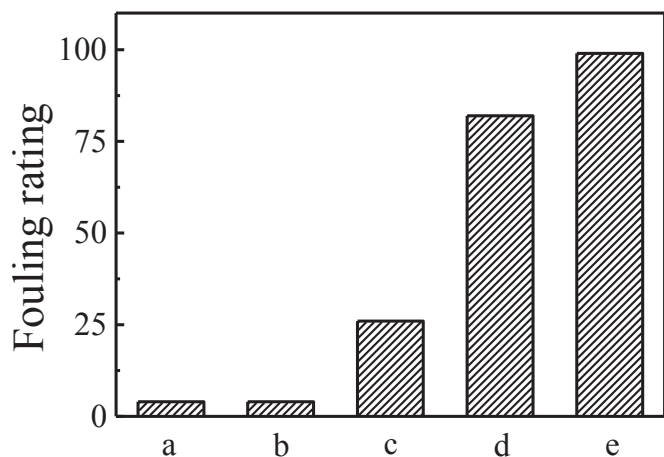


Fig. 7. The fouling rating after immersion in the sea for four months. (a) control panel without coating, (b) panel coated with PCL, (c) panel coated with PCL/PBS = 80/20, (d) panel coated with PCL/DCOIT = 90/10, (e) panel coated with PCL/PBS/DCOIT = 72/18/10.

Acknowledgments

The financial support of Ministry of Science and Technology of China (2012CB933802), National Natural Science Foundation of China (51573061, 51303059), Natural Science Foundation of Guangdong Province (S2013040016085), and the Fundamental

Research Funds for Central Universities (2014ZZ0003) is acknowledged.

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